

# Relation Algebra

§9.  $\frac{R}{S}$

$$R \div_s S \triangleq (R \div S) \cap (S \div R)^\circ$$

what  $R$  sends  $x$  to and nothing else is what  $S$  sends  $y$  to. In Rel  $x$  and  $y$  has the same image:  $\forall p. (x R p \iff y S p)$

(9a)

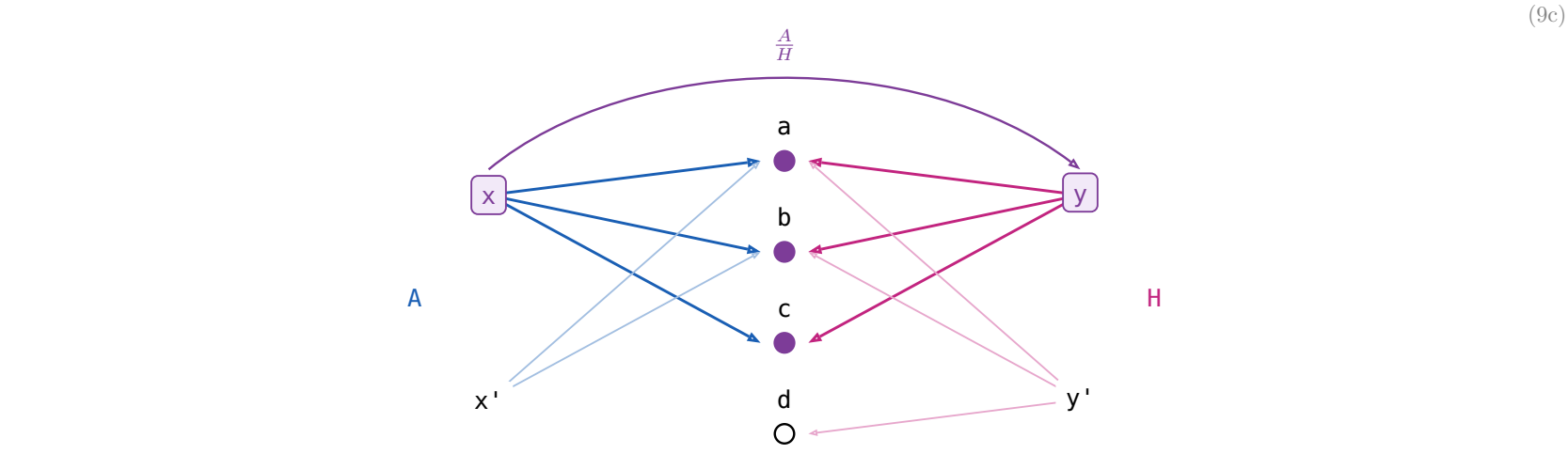
$$x \div (A/H) y \text{ --- } x \text{ admires everyone } y \text{ hates}$$

$$x \div ((H/A)^\circ) y \text{ --- } x \text{ admires only people } y \text{ hates}$$

$$x \div ((A/H) \cap (H/A)^\circ) y \text{ --- } x \text{ admires only and all whom } y \text{ hates}$$

(9b)

$x$  admires exactly whom  $y$  hates:  $\div$  is **all of**,  $\frac{R}{S}$  is **only all of**.



| law                                                                                                                                                                                                    | the reading                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \sqsubseteq R /_s S \iff X S \sqsubseteq R \wedge X^\circ R \sqsubseteq S$<br>$X$ sits inside the matching exactly when it carries $S$ back inside $R$ and, reversed, carries $R$ back inside $S$ . | $X$ may pair $x$ with $y$ only when $x$ admires exactly whom $y$ hates. Both halves must typecheck, so the operation is <b>partial</b> .              |
| $(R /_s S)^\circ = S /_s R$ reversing the matching swaps the two relations.                                                                                                                            | Matching is symmetric.                                                                                                                                |
| $(R /_s S) (S /_s W) \sqsubseteq R /_s W$ a match from $R$ to $S$ followed by one from $S$ to $W$ is a match from $R$ to $W$ .                                                                         | And transitive.                                                                                                                                       |
| $(R /_s S) S \sqsubseteq R$ following the matching by $S$ lands inside $R$ .                                                                                                                           | $(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$<br>$x$ only admires whom $y$ hates<br>$\frac{R}{S}S = \text{Dom}(\frac{R}{S})R$             |
| $(R /_s R) R = R$ matching $R$ against itself and then following $R$ gives $R$ back.                                                                                                                   | $(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$<br>$x$ and $y$ admire the same people<br>$y=x$ always qualifies ( $1 \sqsubseteq R \circ R$ below) |
| $1 \sqsubseteq R /_s R$ every $x$ matches itself.                                                                                                                                                      | $x(\frac{R}{R})y$ if $x$ and $y$ admires the same peoples.                                                                                            |
| $(R /_s R) (R /_s R) = R /_s R$ matching twice matches no more pairs than matching once.                                                                                                               | So the relation <b>admires the same people</b> is an equivalence relation.                                                                            |
| $\text{Symmetric}(X) \implies X \sqsubseteq R /_s R \iff X R \sqsubseteq R$ a symmetric $X$ sits inside the matching exactly when following it by $R$ adds nothing to $R$ .                            | The largest symmetric arrow that leaves $R$ alone.                                                                                                    |
| $\text{simplePart}(R) \triangleq R /_s 1$ matching $R$ against the identity keeps the $x$ whose image is a single point.                                                                               | The people who admire exactly one person and nobody else. It equals $R$ only when $R$ is simple, unlike $R/1=R$ .                                     |
| $\text{dom}(R /_s S) = 1 \cap (R /_s S) (S /_s R)$<br>the $x$ that match something are those the two one-sided divisions take round the loop back to $x$ .                                             | Its domain is the <b>domain of simplicity</b> of $R$ .                                                                                                |